NEXT SIGNAL

Track D: Pharma and Healthcare | Team IdeaForge
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THE APPROACH IN ONE LINE

Next Signal computes before it reads, looks outside the company before it trusts the company, and spends most of its effort trying to kill its own hypotheses. A discovery is whatever survives that effort.

1. Why this is not a summariser

A system that only reads a company's own documents mostly restates what management already chose to emphasise. The brief's own worked example finds its story in a **computed** gap between narrative (volume-led premium growth) and arithmetic (price versus volume split, receivable days rising from 28 to 46). A summarising agent that does not compute this from raw disclosed numbers cannot reach that kind of finding, because the finding only exists after the calculation. Track D compounds this: the most informative evidence often sits outside the company entirely (USFDA inspection records, a manufacturing partner's regulatory status, peer segment numbers for the same quarter), which a single-corpus retrieval agent has no mechanism to go find or to notice contradicts the company's own framing.

Our build inverts the usual order. A deterministic metric library (44 formulas) computes peer z-scores and statistical change-points across every disclosed period *before* any language model is asked to form an opinion. The model's only job is to explain an anomaly the arithmetic has already found, never to originate a number shown to the user.

2. Tools and stack

Layer	Choice	Why
Research LLM	Sarvam `sarvam-105b`, JSON-schema structured output, Gemini fallback on 429/5xx	128K context, OpenAI-compatible tool calling, India-first provider
Voice analyst	Sarvam `sarvam-105b-conversations` + `saaras:v3` STT + `bulbul:v3` TTS	Grounded only on the generated report; never invents a number
Orchestration	Python FastAPI, async stage pipeline, SQLite run store, SSE event bus	Six named stages, each independently testable, no agent framework needed for a fixed pipeline
Retrieval	BM25 over PDF pages (rank_bm25), quote-anchored extraction	Deterministic, auditable, no vector database or embedding drift
Statistics	ruptures (PELT change-points), peer z-scores, event-study (market model, CAR)	The same methods real analysts use for channel checks and earnings-quality screens
Data sources	NSE/BSE filings, openFDA, FDA warning letters and Import Alert 66-40, Orange Book, SEC EDGAR (for ADR issuers),	Public only; the exact chain the brief's evidence-trail diagram implies

4. Track D, and generalisation without tuning

The universe is read live from NSE Indices CSVs at run time, never hard-coded: NIFTY 50 filtered to Healthcare (5 names today) plus the tie-break universe (Nifty Pharma and Nifty Healthcare minus those 5, roughly 19 further companies as of October 2026). A company routes to one of two sub-industry playbooks, pharma (10 angles: business- model shift, regulatory-before-mainstream, geographic mix, exclusivity cliffs, launch-masks-erosion, working capital, R&D intensity, profit quality, partner dependence, cross-channel disclosure) or hospital (6 angles: announced-versus-commissioned beds, ARPOB decomposition, occupancy dilution, segment drag, regulatory price exposure, organic-versus-acquired growth), plus a generic fallback (6 angles) for any company outside those two. A company-name test asserts no symbol or company name appears anywhere in the engine's own logic, only in the universe loader's live data and the evaluation harness.

5. Triangulation and contradiction detection

Evidence is counted by independent *origin*, not by document: a press item that merely restates a company filing within three days collapses into that filing's origin rather than counting as a second source. Five contradiction types are checked automatically: a numeric mismatch between two sources for the same metric and period (primary-source precedence resolves it); a narrative-versus-number divergence (management's framing points one way, the computed trend points the other); a guidance-versus-actual miss; a cross-exchange mismatch (NSE versus SEC); and an aggregator-versus-primary disagreement. This last type is not theoretical: while building this system, two secondary sources gave materially different net-profit figures for the same company and quarter, and the primary filing (opened directly) confirmed one was simply wrong. The engine's rule, applied here by hand before it was automated, is to always trust the primary filing.

6. Self-falsification

Every hypothesis faces the same nine-test battery before it can become a discovery. A hypothesis that only explains a sector-wide move, or that disappears once a disclosed one-off is excluded, or that a rival explanation scores more consistently in the Analysis of Competing Hypotheses matrix, is downgraded or killed, and the kill reason is shown, not hidden. The Falsification Condition itself is built from the hypothesis's own first prediction: a named metric, a comparator, a threshold derived from the pre-change period or the peer median, and a dated next check against

	GDELT, CRISIL/ICRA/CARE, NSE MCP, Angel One, Upstox	
UI	Next.js 16, TypeScript, Tailwind v4, recharts, framer-motion	Click-through evidence drawer on every number
Exports	Markdown, HTML-to-PDF (headless Chrome), XLSX (openpyxl), CSV bundle	Report and dashboard both required by the brief; we build both

3. Workflow: ticker to tested discovery

01 Discover	02 Connect	03 Calculate	04 Reason	05 Challenge	06 Prove

1. **Discover.** Resolve the company (NSE index CSVs, live if needed), score candidate angles by `0.35·data_availability + 0.40·anomaly_signal + 0.25·novelty_prior`, keep the top 4.
2. **Connect.** Fetch documents in parallel; BM25 ranks candidate pages per metric; an LLM extracts facts with a verbatim quote, verified character-for-character against the source page before it is trusted.
3. **Calculate.** 44 formulas (growth, mix, profitability, working capital, regulatory, hospital metrics), peer-relative z-scores, PELT change-point detection, and an event study against NIFTY 50.
4. **Reason.** An LLM proposes up to 6 hypotheses from the ranked anomalies, each with explicit predictions and disconfirmers; deterministic rival hypotheses (sector, FX, one-off, accounting) are generated automatically.
5. **Challenge.** Nine falsification tests (peer, FX, one-off, reclassification, seasonality, persistence, novelty, a disconfirming search, a point-in-time replay) plus an Analysis of Competing Hypotheses matrix. The strongest surviving rival becomes the stated Counterargument.
6. **Prove.** A deterministic strength score (0 to 10: independence, reliability, corroboration, survival, persistence, minus penalties) labels the discovery Strong, Moderate or Weak with its reasoning shown, builds the Falsification Condition from the hypothesis's own prediction, and writes the brief's eight required elements.

a named future filing. If nothing survives, the system says so plainly rather than forcing a weak claim.

7. Output: dashboard and report, both

A Next.js dashboard shows the discovery, a strength bar with the Moderate/Strong thresholds marked, a swimlane timeline with the change-point drawn on the plot, an Analysis of Competing Hypotheses heat grid, and an evidence drawer that opens the exact quote and source for any number clicked. The same run exports to Markdown, a styled PDF, an Excel workbook (one sheet per table), and a CSV bundle, so either required format (report or dashboard) is simply a button click from the other.

A compact illustration of the eight-element format, built from the engine's own schema and not yet a live run's output, is given in the appendix below.

8. Build plan and milestones

Date	Milestone	Status
11 to 14 Oct	Research deliverable: 10 lanes, ~830 sources, company dossiers for all 5 NIFTY 50 names plus 19 tie-break companies	Done
11 to 15 Oct	Engine core: schemas, HTTP/cache, universe loader, 44-metric library, change-point/event-study, Admiralty grading, independence counting, novelty scoring, 6 of 9 falsification tests, ACH matrix, strength scorer, LLM router with fallback, guards, full export chain (MD/HTML/PDF/XLSX/CSV, PDF verified against a real headless-Chrome render)	Done, 126 tests passing
15 Oct	Dashboard: home, live-run page (SSE-driven stage rail and hypothesis board), report page (discovery hero, evidence drawer, ACH matrix, challenge tab, exports), eval and about pages	Done
16 to 17 Oct	Pipeline wiring: Discover through Prove stages connected to real sources, voice analyst ported and grounded on the live report	In progress
17 to 18 Oct	Evaluation harness across all 5 NIFTY 50 names plus the tie-break universe; time-travel and ablation runs	Scheduled
19 Oct	Round 1 submission	This document
20 to 24 Oct	Figma deck, rehearsal, hardening	Scheduled
25 Oct	Finals: live run on a judge-chosen company	Scheduled